Python core programming

Units 1–5 · syllabus 2024, Calicut University BSc Physics Honours · module marks 12, doubled to 24

HOW THIS PAPER WORKS

The syllabus gives this module 12 marks. This paper gives **twice** that (24 marks), so every unit is tested with more questions than a real paper would use.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match. Each question also shows its unit (**U1** to **U5**). Every concept of units 1–5 is the main point of at least one question.

Similar papers used: CSC3MN206 Programming using Python, Nov 2025, versions A and B (**N25-A**, **N25-B**); MAT4CJ205 Fundamentals of Python and SageMath, Apr 2026 (**A26**).

Time 1 hour 30 minutes
Maximum 24 marks

Section A: answer all. Section B: answer any two of three. Write the Python code exactly as it would run in Jupyter or Colab.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 16.

1. Name the three numerical data types in Python. Give one example of each, and state which one is written with a letter `j`.

PYQ N25-B Q3 · adapted · U1

2. What is printed by the following code? Explain in one line why.

```
x = [1, 2, 3]
x.append(4)
print(x[1:3], len(x))
```

Instructor · U1

3. State two differences between a list and a tuple. Why can a tuple not be changed after it is created?

PYQ N25-B Q18 · adapted · U2

4. A student stores the pendulum period for each string length in a dictionary. Write one line of Python that creates the dictionary `T = {0.25: 1.00, 0.50: 1.42, 1.00: 2.01}` and one line that prints the period for length 0.50.

PYQ N25-A Q8 · adapted · U2

5. Write the code that opens a text file named `data.txt` for reading, prints every line, and closes the file. Name the function used to open the file and the method used to close it.

Instructor · U2

6. What does the following loop print? Explain the role of `break` in it.

```
for i in range(10):
    if i % 3 == 0:
        break
    print(i)
```

Instructor · U3

7. Write a loop that prints the odd numbers from 1 to 9, skipping the number 5 with `continue`.

Instructor · U3

8. Write one line of NumPy code that creates a 3 × 3 array of random integers from 1 to 10 (both included). Explain why the value written as 11 in the code is not included in the range.

PYQ A26 Q12 · adapted · U5

Section B · Descriptive

Answer any two of the three. Each carries 4 marks. Total 8.

9. Write a function `extremes(values)` that accepts a list of numbers and returns the largest and smallest numbers as a tuple. Explain what is returned, and state the output for `extremes([4, 9, -2, 7])`.

PYQ N25-A Q7 · adapted · U4

10. Two 3 × 3 NumPy arrays `A` and `B` are created. Explain how `A + B`, `A - B` and `A * B` work element by element. Then explain what happens when `A / B` is computed and one element of `B` is zero.

PYQ N25-B Q6 · adapted · U5

11. A student measures the period of a simple pendulum five times and stores the readings in a list `periods = [2.01, 1.98, 2.03, 2.00, 1.99]`. Write a function that computes the mean of the list using a loop and returns it, then print the mean rounded to two decimal places.

Instructor · U1, U4

COVERAGE

UNIT	TOPIC	QUESTIONS
1	Variables, operators, int / float / complex, lists	1, 2, 11
2	Tuples, sets, dictionary, input, open() and close()	3, 4, 5
3	if / else, break and continue	6, 7
4	Functions: arguments and return values	9, 11
5	NumPy arrays: creation, access, operations	8, 10

Every unit is covered. Total marks: 16 (Section A) + 8 (Section B) = 24.

Pandas DataFrame

Units 6–10 · syllabus 2024 · module marks 15, doubled to 30

HOW THIS PAPER WORKS

The syllabus gives this module 15 marks. This paper gives **twice** that (30 marks). The questions use the physics data named in the syllabus: capacitor charging, phone-sensor readings, and pendulum and spring experiments.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: CSC1MN106 Computer Fundamentals with MS Excel, SPSS (Nov 2024); STA2MN101 Probability Theory I (Apr 2025); STA3MN201 Statistical Inference using R (Nov 2025); MAT4CJ205 Python and SageMath (Apr 2026).

Time 2 hours
Maximum 30 marks

Section A: answer all ten. Section B: answer any two of three. Show the code you would run. Where a table is given, write the output as a table.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 20.

1. Write the code that creates a DataFrame `df` with two columns, `Length` and `Period`, from the lists `[0.25, 0.50, 1.00]` and `[1.00, 1.42, 2.01]`.
PYQ A26 Q16 · adapted · U6

2. For the DataFrame of question 1, state what `df.shape`, `df.columns` and `df.dtypes` return.
Instructor · U7

3. Long-format data has columns `trial`, `capacitor` and `voltage`. Explain what `df.pivot(index='trial', columns='capacitor', values='voltage')` produces.
PYQ CSC1MN106 N24 A7 · adapted · U7

4. What is the difference between `df.sort_values('Period')` and `df.sort_index()`? Which one sorts by row labels?
Instructor · U7

5. A sensor file has the missing value `-999` in some rows. Write one line that replaces every `-999` with `NaN`, and one line that drops every row that still contains a `NaN`.
Instructor · U8

6. Explain what `df.describe()` reports for a numeric column, and state what `df['Period'].idxmax()` returns.
PYQ CSC1MN106 N24 B18 · adapted · U8

7. Write the code for the mode and the count of non-missing values in a column named `mass`.
PYQ STA3MN201 N25 B13 · adapted · U9

8. A student computes the covariance and the correlation of mass and extension for a spring experiment. State the range of the correlation coefficient and say which of the two quantities has units.
PYQ STA2MN101 A25 B15 · adapted · U9

9. Explain what `df['Period'].quantile(0.5)` returns, and write one line that applies `np.sqrt` to every value in a column `Period` using `apply()`.
Instructor · U9

10. A table has a column `material` with values `steel`, `copper`, `rubber` and a column `k`. Write the code that finds the mean value of `k` for each material using `groupby()`.
Instructor · U10

Section B · Descriptive

Answer any two of the three. Each carries 5 marks. Total 10.

11. Capacitor charging. Three capacitors are charged and the voltage is logged every second. The data is read from `charge.csv` with columns `time`, `C1`, `C2` and `C3`, and some cells are empty. Write the code that (a) reads the file, (b) fills the empty cells with the column mean, and (c) prints the correlation between `time` and `C1`. Explain why the correlation alone does not show that the voltage follows an exponential law.

PYQ CSC1MN106 N24 B12 · adapted · U8, U17

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12. Two tables are given. `students` has columns `name` and `group`; `marks` has columns `name` and `score`. Explain the difference between `pd.merge(students, marks, on='name')` and `pd.concat([students, marks], axis=1)`. Give a short example where the two give different results.

PYQ CSC1MN106 N24 A8 · adapted · U10

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13. A phone sensor records acceleration once every 0.1 second for one minute. Write code that (a) builds a time series with `pd.date_range` using the correct frequency, (b) makes it a Series indexed by time, and (c) selects the readings between the 10th and the 20th second. Explain the role of the index in step (c).

Instructor · U10

COVERAGE

UNIT	TOPIC	QUESTIONS
6	Create a DataFrame	1
7	Attributes, pivot, sort, sort by labels	2, 3, 4
8	Missing data, combining, describe, min and max index	5, 6, 11
9	count, mode, covariance, correlation, quantiles, pipe, apply	7, 8, 9, 11
10	Aggregation, groupby, merging, concatenating, time series	10, 12, 13

Total marks: 20 (Section A) + 10 (Section B) = 30. Every topic of units 6–10 is the main point of at least one question. **pipe()** is asked in Section A, question 9.

Visualisation tools: Pandas and Seaborn

Units 11–16 · syllabus 2024 · module marks 15, doubled to 30

HOW THIS PAPER WORKS

The syllabus gives this module 15 marks. This paper gives **twice** that (30 marks). Plots are drawn from the physics data named in the syllabus: capacitors, the convex-lens liquid-lens experiment, the sonometer, and the Iris data set.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: MAT4CJ205 Python and SageMath (Apr 2026); STA3MN201 Statistical Inference using R (Nov 2025); STA2MN101 Probability Theory I (Apr 2025).

Time 2 hours
Maximum 30 marks

Section A: answer all ten. Section B: answer any two of three. Where a question asks for a plot, write the code and say what the plot shows.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 20.

1. Give two reasons why a graph is more useful than a table of readings for finding a physical law. Readings taken at regular times (for example a capacitor voltage every second) are best shown as a time series. Write one line of seaborn code that draws `voltage` against `time` from a DataFrame `df`.

PYQ A26 Q4 · adapted · U11, U16

2. Write the code that draws a bar chart of the mean voltage of three capacitors `C1, C2, C3` stored in a Series named `means`, with the title "Mean voltage" and axis labels.

PYQ STA3MN201 N25 A1 · adapted · U11

3. A histogram of 200 readings of a pendulum period is drawn with `bins=10` and then with `bins=50`. Explain how the shape of the plot changes, and which choice is more honest about the spread of the data.

Instructor · U12

4. What does a box plot show? Name the five numbers it summarises, and state what the length of the box represents.

Instructor · U12

5. Write the code for a scatter plot of `length` against `period` with red circles of size 40 and the title "Pendulum". Name the keyword that sets the marker shape and the keyword that sets the marker colour.

PYQ A26 B14 · adapted · U12

6. Explain how a correlation matrix plot is made from a DataFrame, and state what a value close to +1 in the matrix means for two columns `mass` and `length`.

PYQ STA2MN101 A25 B15 · adapted · U13

7. What does `sns.set_palette('pastel')` change, and why would a physics student pick a colour palette that is safe for colour-blind readers?

Instructor · U14

8. State the difference between a histogram and a kernel density plot. Which one is a smooth curve, and what does its area represent?

Instructor · U14, U15

9. Write the code that draws a violin plot of `petal_length` for each `species` in the Iris data set, using a DataFrame `iris`.

Instructor · U15

10. Explain what a hexbin plot shows and when it is preferred to a scatter plot. Then explain how a bubble chart differs from a scatter plot, and write one line of matplotlib code that draws it from arrays `x`, `y` and a size array `s`.

Instructor · U15, U16

Section B · Descriptive

Answer any two of the three. Each carries 5 marks. Total 10.

11. Liquid-lens experiment. Five readings of the refractive index of a convex lens filled with liquid are stored in a DataFrame with columns `trial` and `n`. Write code that draws a box plot of `n`, adds a title and axis labels, and saves the figure as `lens.png`. Then explain what an outlier in a box plot is, and how you would recognise one in this data.

Instructor · U12

12. Sonometer. The mass `M` and the length `L` are stored in the DataFrame `son`. Write code that (a) adds a column `L2 = L**2`, (b) draws a regression plot of `M` against `L2` with `sns.regplot`, and (c) prints the correlation matrix of `M` and `L2`. Explain what the slope of the line would mean physically.

Instructor · U13, U16

13. Iris data. Using the Iris set, write code that draws a pair grid of the four numerical columns, coloured by `species`, and a heatmap of the correlation matrix of the same four columns with the colour map `'coolwarm'`. Explain what a pair grid shows that a single histogram does not.

Instructor · U15, U16

COVERAGE

UNIT	TOPIC	QUESTIONS
11	Importance of visualisation, bar chart	1, 2
12	Histogram, frequency polygon, box plot, scatter plot, markers, labels, title	3, 4, 5, 11
13	Correlation matrix and correlation plot	6, 12
14	Seaborn features, colour palette, univariate distribution plot	7, 8
15	Seaborn histogram, density, bivariate plot, hexbin, violin, pair grid	8, 9, 10, 13
16	Bar plot for categories, regplot, heatmap, cmap, bubble chart, time series	1, 10, 12, 13

Total marks: 20 (Section A) + 10 (Section B) = 30. Every syllabus topic of units 11–16 is the main point of at least one question.

Data file formats and Jupyter

Units 17–19 · syllabus 2024 · module marks 8, doubled to 16

HOW THIS PAPER WORKS

The syllabus gives this module 8 marks. This paper gives **twice** that (16 marks). Every file format named in unit 17 (csv, xls, tab, dat) is tested.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: MAT4CJ205 Python and SageMath (Apr 2026); CSC1MN106 Computer Fundamentals with MS Excel, SPSS (Nov 2024).

Time 1 hour

Maximum 16 marks

Section A: answer all six. Section B: answer one of two. Assume the files are in the same folder as the notebook.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 12.

1. What is the difference between a Series and a DataFrame in pandas? Give one example of each.

PYQ A26 B16 · adapted · U17

2. A capacitor experiment is saved as a comma-separated file `cap.csv`. Write the one line of code that loads it into a DataFrame `df`.

PYQ A26 Q6 · adapted · U17

3. A phone sensor file `accel.tab` is separated by tab characters. Write the code that loads it into `df`, and explain the argument you must pass to `pd.read_csv` or `pd.read_table` to make the tabs work.

Instructor · U17

4. Name the function used to read an Excel `.xls` or `.xlsx` file. What must be installed on the machine first if the file is an `.xls` from an older Excel version?

PYQ CSC1MN106 N24 B12 · adapted · U17

5. A plain-text data file `spring.dat` has values separated by spaces. Explain why `sep=' '` or `sep=r'\s+'` is used when reading it, and state what happens if the default comma separator is kept.

Instructor · U17

6. Given `df` with a row index of 0, 1, 2, 3, write the code for (a) the value in row 2, column `'Period'` using `loc`, and (b) the first two rows and the first column using `iloc`. Say in one line how `loc` and `iloc` differ.

Instructor · U18

Section B · Descriptive

Answer one of the two. Carries 4 marks. Total 4.

7. Explain the work environments: what Jupyter Notebook is, how Anaconda provides it, and how Google Colab differs from a local install. Name one reason a physics student would prefer Colab.

Instructor · U19

8. A student has three files for the same experiment: `run1.csv`, `run2.dat` and `run3.tab`, with columns `length` and `period`. Write a loop that reads all three into one list of DataFrames, then uses `pd.concat` to join them, and explain why the separators are different in each file.

Instructor · U17, U18

COVERAGE

UNIT	TOPIC	QUESTIONS
17	Series and DataFrames, csv, xls, tab, dat formats	1, 2, 3, 4, 5, 8
18	loc and iloc, operations on DataFrames	6, 8
19	Jupyter Notebook, Anaconda, Google Colab	7

Total marks: 12 (Section A) + 4 (Section B) = 16.

Open-ended module: additional training on data analysis

Examples 1–5 of the syllabus · module marks 5, doubled to 10

HOW THIS PAPER WORKS

The syllabus gives the open-ended module 5 marks. This paper gives **twice** that (10 marks). Each example from the syllabus appears once. You answer one example from each part.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: CSC1MN106 Computer Fundamentals with MS Excel, SPSS (Nov 2024); STA2MN101 Probability Theory I (Apr 2025).

Time 1 hour

Maximum 10 marks

Write complete programs. Each program must read or create its data, process it, and show a plot or print a result. Use the labels and file names given in the question.

Part A · Files and plotting

Answer one of the two. Each carries 5 marks. Total 5.

1. Data file creation (example 1). Write a Python program that generates a CSV file `pendulum.csv` with two columns, `Length` and `Period`, for lengths 0.25 m, 0.50 m, 0.75 m and 1.00 m. Use three period readings per length (values of your own choosing, close to the theory $T = 2\pi\sqrt{L/g}$). Build the data as a pandas DataFrame and save it with `to_csv`, without the index column.

Instructor · Example 1

2. File read and plot (example 2). Write a Python program that reads `pendulum.csv` (the file from example 1), computes the mean period for each length using `groupby()`, and uses Seaborn to draw a regression line of mean period against length. Label the axes and give the plot a title. Write one sentence on what the plot says about the relation between period and length.

Instructor · Example 2

Part B · Merging and visualisation

Answer one of the two. Each carries 5 marks. Total 5.

3. Merge and group (example 3). Two DataFrames are given. `springs` has columns `spring_id`, `material` and `force`, `extension`. `materials` has columns `material` and `youngs_modulus`. Write code that (a) merges the two tables on `material`, (b) computes the spring constant `k = force / extension` for each row, and (c) finds the mean spring constant for each material using `groupby()`. Explain why the merge step is needed before the group step.

PYQ CSC1MN106 N24 A8 · adapted · Example 3

4. Seaborn visualisation (example 4). Using the table from example 3, draw the linear relationship between the force applied and the extension with `sns.regplot`. Also draw, in the same program, a pandas histogram of the extension column and a pandas scatter plot of force against extension. Explain in two sentences how the regression line shows Hooke's law.

PYQ STA2MN101 A25 B16 · adapted · Example 4

COVERAGE

SYLLABUS ITEM	QUESTION
Example 1: data file creation with a DataFrame	1
Example 2: read the file, mean period, regression line	2
Example 3: merge and group by, spring constant with material	3
Example 4: regplot of force against extension	4
Pandas histogram and scatter plot	4
Seaborn visualisation tools	2, 4

Total marks: 5 (Part A) + 5 (Part B) = 10. Bar charts in pandas are covered in Module 3, question 2.